Chapter 6: Network Topologies, Architectures & Types

A Study Guide for Aspiring Network Professionals

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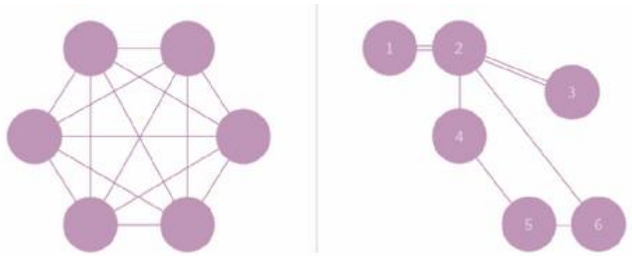
Chapter 6 — N10-009 Exam Objective Deep Dive

> Chapter 6 of the Network+ N10-009 exam asks you to compare and contrast network topologies, architectures, and types. This means knowing not just what each one is, but when to use it, what its weaknesses are, and how to recognize it on the exam. This post covers all nine exam objectives in full.

1. Mesh Topology

In a mesh topology, every device connects to every other device — either fully or partially. There are two variants you must know:



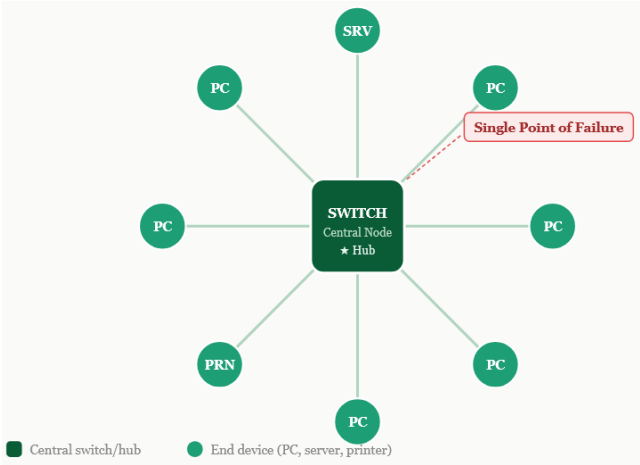
Full mesh	Partial mesh
• Every node connects to every other • Maximum redundancy • Expensive at scale • Requires n(n−1)/2 links	• Only critical nodes are fully interconnected • Cost-effective compromise • Common in WANs and ISP backbones

Exam tip:

*Mesh is used where **fault tolerance** is paramount — think WAN links, military networks, and the internet itself. A single link failure does not bring down the network.*

2. Star / Hub-and-Spoke Topology

The most common modern LAN topology. Every end device connects to a central switch or hub. All traffic passes through that central device.



Advantages	Disadvantages
• Easy to troubleshoot • Single node failure doesn't affect others • Simple to add/remove devices	• Central switch = single point of failure • Performance limited by the central device

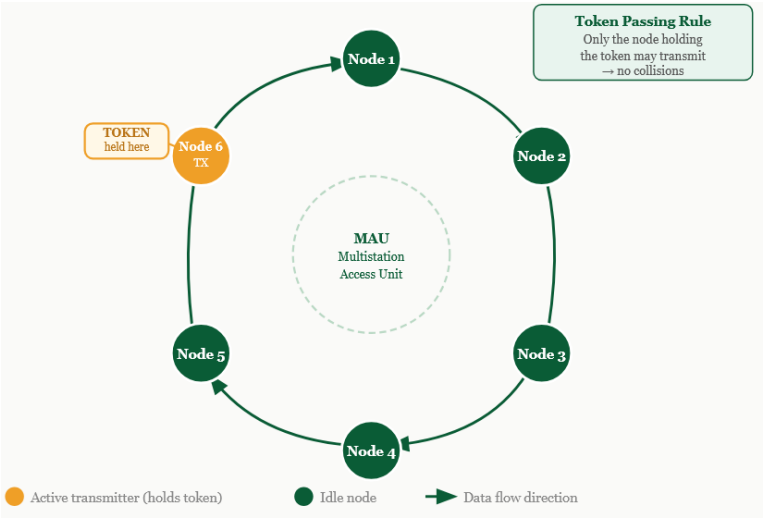
Hub vs. switch:

Older star networks used hubs (half-duplex, shared bandwidth). Modern networks use switches (full-duplex, dedicated bandwidth per port). On the exam, "star" almost always implies a switch today.

3. Token Ring Topology

A legacy topology (IEEE 802.5) where devices form a logical ring and pass a special frame called a token. Only the device holding the token may transmit — this eliminates collisions without needing CSMA/CD.

Devices are wired to a central Multistation Access Unit (MAU), making the physical layout look like a star, while the logical data flow is a ring.

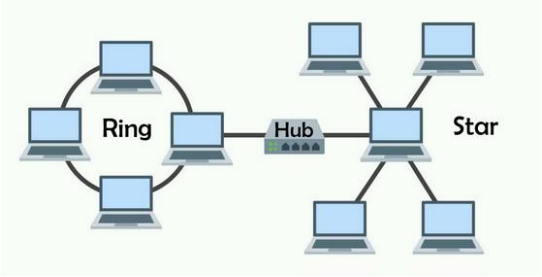


Why it still appears on the exam:

Network+ tests your understanding of how collision domains were historically managed. Token ring is essentially retired, but knowing how token passing avoids collisions is conceptually important.

4. Hybrid Topology

A hybrid topology combines two or more different topology types into a single network. Real-world enterprise networks are almost always hybrid.



Common examples:

- **Star-bus:** Multiple star LANs connected via a backbone bus
- **Star-mesh:** Star LANs at branch offices, full-mesh WAN links between sites
- **Star-ring:** Star physical layout with token ring logical flow

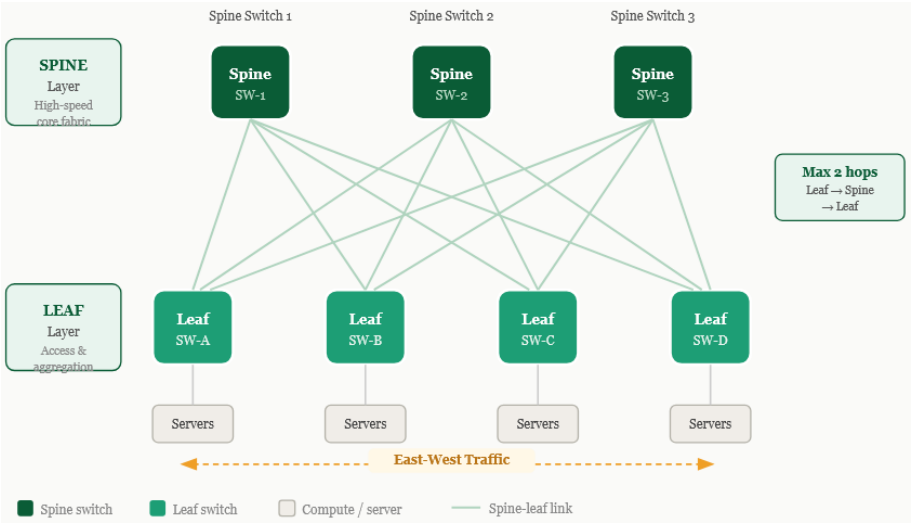
Exam angle:

If a question describes multiple topology characteristics in a single network, it's describing a hybrid. Recognize the components — don't get confused into thinking it's a new topology.

5. Spine-and-Leaf Architecture

A modern data center architecture designed for predictable, low-latency east-west traffic (server-to-server). It replaces the older three-tier model in hyperscale environments.

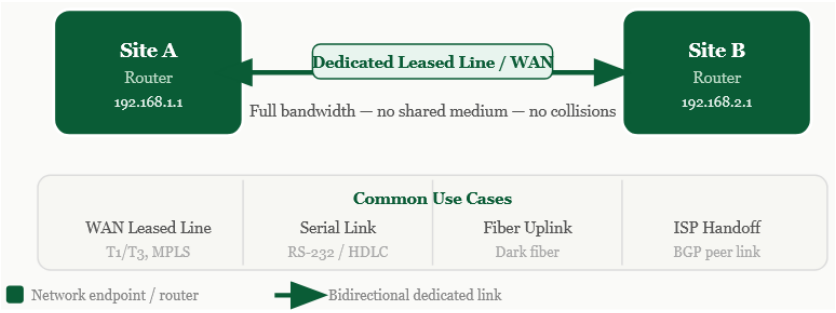
- **Leaf switches:** Connect directly to servers, storage, and firewalls
- **Spine switches:** Interconnect all leaf switches — no leaf-to-leaf direct link
- Every leaf connects to every spine (partial mesh between layers)
- Traffic between two servers crosses exactly two hops: leaf → spine → leaf



Strengths	Considerations
• Consistent, predictable latency • Easy horizontal scaling • No spanning tree needed	• More cabling than three-tier • All spine switches must be high-capacity

6. Point-to-Point Topology

The simplest possible topology: two nodes connected directly with a dedicated link. Think of a leased line between two offices, a serial WAN link, or a direct fiber run.



Key characteristics:

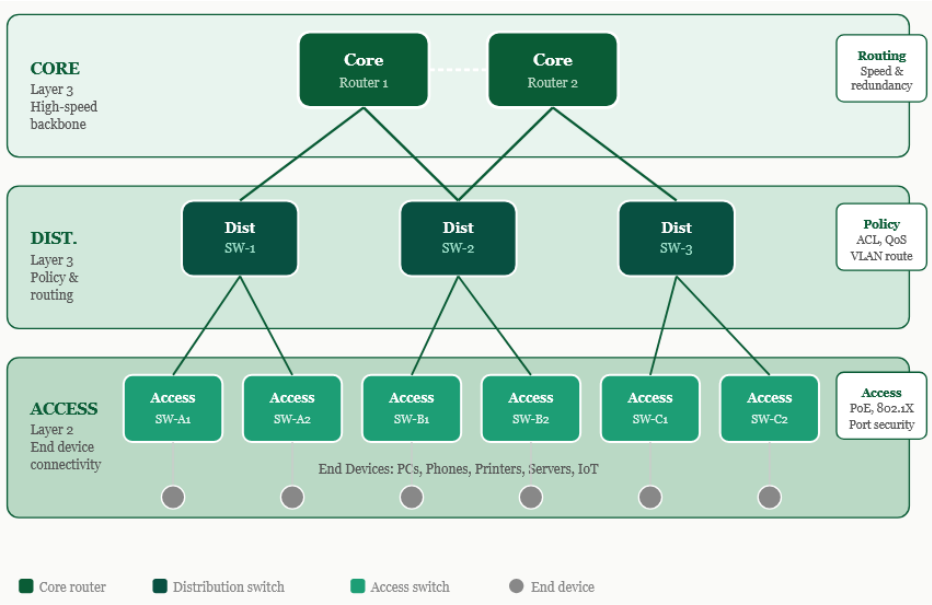
- No shared medium — full bandwidth dedicated to the link
- No collision domain concerns
- Common for WAN uplinks, ISP handoffs, and backbone connections

Don’t confuse:

Point-to-point with point-to-multipoint (like a hub broadcasting to many endpoints). P2P = exactly two endpoints, one dedicated path.

7. Three-Tier Hierarchical Model

The classic Cisco enterprise campus architecture. Three distinct layers, each with a defined role:



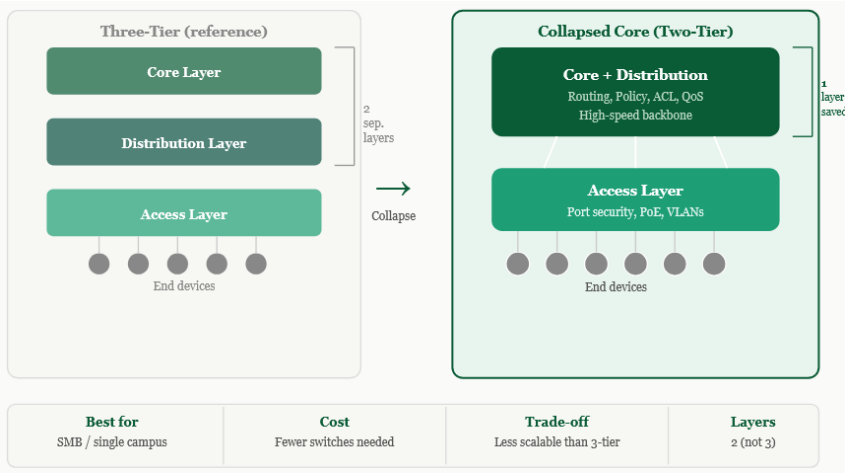
Core Layer	High-speed backbone. Routes packets as fast as possible. No policy enforcement — speed is everything. Think: aggregated fiber, high-end routers.
Distribution Layer	Policy enforcement, routing between VLANs, QoS, access control lists, and summarization. Acts as the boundary between access and core layers.
Access Layer	Where end devices connect. Switches provide port-level security (802.1X, port security), PoE, and VLAN assignment. This is where you'll find users, printers, and phones.

Mnemonic:

Core, Distribution, Access → C-D-A → "Can't Do Anything" — each layer has a specific job; policy lives in the middle.

8. Collapsed Core

When a network is small enough that a separate core layer isn't justified, the core and distribution layers are merged into a single layer. This is called **collapsed core** (or ***two-tier***) architecture.



When to use it	Trade-offs
• Small/medium enterprise • Single-building campus • Budget constraints	• Less scalable • Core devices must handle policy too • Harder to upgrade without downtime

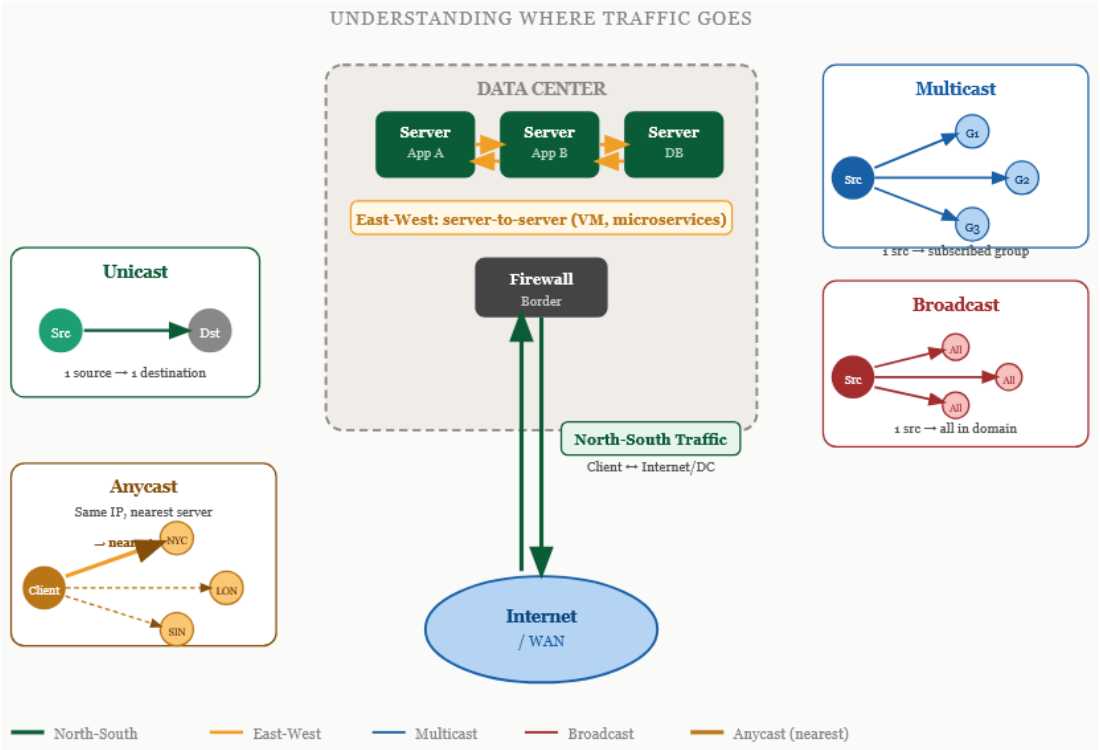
Exam tip:

A question describing a two-layer campus design (access + core/distribution combined) is describing a collapsed core. It's a cost-justified simplification, not a flaw.

9. Traffic Flows

Understanding where traffic goes is as important as understanding physical topology. Network+ N10-009 expects you to know these primary traffic patterns:

North-South	Traffic between clients (inside) and servers or the internet (outside). Flows vertically — in and out of the data center. Traditional firewall policies focus here.
East-West	Traffic between servers within the data center — VM-to-VM, microservice-to-microservice. Modern workloads are dominated by east-west traffic, hence spine-and-leaf designs.
Unicast	One source to one specific destination. Standard day-to-day traffic. Most HTTP/HTTPS, SSH, and file transfer sessions are unicast.
Multicast	One source to a subscribed group of receivers. Used for video streaming, routing protocols (OSPF, EIGRP), and IPTV deployments.
Broadcast	One source to all devices in a broadcast domain. ARP requests are the classic example. Switches forward broadcasts; routers block them.
Anycast	One source, routed to the nearest of many possible receivers. Core to how DNS root servers and CDNs work — same IP address advertised from multiple global locations.



Why traffic flows matter for topology choice:

A spine-and-leaf architecture exists because east-west traffic exploded with virtualization and microservices. Three-tier models optimized for north-south. Matching your architecture to your traffic flow is a design principle you will see tested.

Chapter 6 — Exam Cheat Sheet

Mesh	Maximum redundancy; full mesh = n(n−1)/2 links; partial mesh common in WANs
Star	Central switch; easiest to troubleshoot; central device is single point of failure
Token Ring	Legacy IEEE 802.5; token passing prevents collisions; MAU at center
Hybrid	Mix of two or more topologies; all real enterprise networks are hybrid
Spine-and-leaf	Data center; 2-hop max between any servers; scales east-west traffic
Point-to-point	Two endpoints, dedicated link, no shared medium
Three-tier	Access → Distribution → Core; distribution = policy layer
Collapsed core	Core + distribution merged; used in smaller networks
Traffic flows	N-S (in/out), E-W (internal), unicast, multicast, broadcast, anycast

Topology questions on the Network+ exam often give you a scenario and ask you to identify the best fit. If you internalize why each topology exists, not just what it looks like, you'll answer those scenario questions confidently. Good luck on N10-009!

Written by a Hassen Hannachi | Part of the Network+ N10-009 Series

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